

Configure IPv6 Addressing

1. Open your Windows or Linux system, use the command line to assign a static IPv6 address to your primary network adapter, and verify the configuration using ipconfig (Windows) or ifconfig /ip (Linux).

ANS:

- **Linux Commands :**

- sudo ip -6 addr add 2001:db8:1::10/64 dev eth0
- ip -6 addr show

- **OutPut :**

```
(root@kali)-[/home/kali]
# sudo ip -6 addr add 2001:db8:1::10/64 dev eth0

(root@kali)-[/home/kali]
# ip -6 addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 state UNKNOWN qlen 1000
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 state UP qlen 1000
    inet6 2001:db8:1::10/64 scope global
        valid_lft forever preferred_lft forever
    inet6 fe80::3d2c:9e04:ad76:6e67/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(root@kali)-[/home/kali]
# ifconfig
docker0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255
    ether fe:ba:c5:28:de:35 txqueuelen 0 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 5 overruns 0 carrier 0 collisions 0

eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.0.107 netmask 255.255.255.0 broadcast 192.168.0.255
    inet6 fe80::3d2c:9e04:ad76:6e67 prefixlen 64 scopeid 0x20<link>
    inet6 2001:db8:1::10 prefixlen 64 scopeid 0x0<global>
    ether 00:0c:29:b6:28:b8 txqueuelen 1000 (Ethernet)
    RX packets 117 bytes 11480 (11.2 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 39 bytes 4920 (4.8 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

2. Configure a unique local IPv6 address (starting with 'fd') on your system and confirm connectivity by pinging another device on your LAN that also has a unique local IPv6 address. Hint: Use the ping6 or ping -6 command for IPv6 testing.

ANS :

```
(root@kali)-[/home/kali]
# ip -6 addr show eth0
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    inet6 fd12:3456:789a:1::10/64 scope global
        valid_lft forever preferred_lft forever
    inet6 2001:db8:1::10/64 scope global
        valid_lft forever preferred_lft forever
    inet6 fe80::3d2c:9e04:ad76:6e67/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(root@kali)-[/home/kali]
# ping6 fd12:3456:789a:1::10
PING fd12:3456:789a:1::10 (fd12:3456:789a:1::10) 56 data bytes
64 bytes from fd12:3456:789a:1::10: icmp_seq=1 ttl=64 time=0.032 ms
64 bytes from fd12:3456:789a:1::10: icmp_seq=2 ttl=64 time=0.041 ms
64 bytes from fd12:3456:789a:1::10: icmp_seq=3 ttl=64 time=0.062 ms
64 bytes from fd12:3456:789a:1::10: icmp_seq=4 ttl=64 time=0.068 ms
64 bytes from fd12:3456:789a:1::10: icmp_seq=5 ttl=64 time=0.065 ms
64 bytes from fd12:3456:789a:1::10: icmp_seq=6 ttl=64 time=0.036 ms
64 bytes from fd12:3456:789a:1::10: icmp_seq=7 ttl=64 time=0.038 ms
64 bytes from fd12:3456:789a:1::10: icmp_seq=8 ttl=64 time=0.053 ms
64 bytes from fd12:3456:789a:1::10: icmp_seq=9 ttl=64 time=0.061 ms
64 bytes from fd12:3456:789a:1::10: icmp_seq=10 ttl=64 time=0.065 ms
64 bytes from fd12:3456:789a:1::10: icmp_seq=120 ttl=64 time=0.070 ms
64 bytes from fd12:3456:789a:1::10: icmp_seq=121 ttl=64 time=0.041 ms
^C
— fd12:3456:789a:1::10 ping statistics —
121 packets transmitted, 121 received, 0% packet loss, time 122884ms
rtt min/avg/max/mdev = 0.032/0.058/0.118/0.014 ms
```

3. Set up a temporary IPv6 address on your system and check if it changes after a reboot or network disconnect/reconnect. Constraint: Document the steps and the observed address behaviour in a text file.

ANS:

1. Checked IPv6 addresses using:

ip -6 addr show dev eth0

```
(kali@kali)-[~]
$ ip -6 addr show dev eth0
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    inet6 fe80::3d2c:9e04:ad76:6e67/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

2. Disconnected network:

```
sudo ip link set eth0 down
```

3. Reconnected network:

```
sudo ip link set eth0 up
```

4. Checked IPv6 addresses again.

```
(kali㉿kali)-[~]
└─$ sudo su
(kali㉿kali)-[/home/kali]
└─# sudo ip link set eth0 down

(kali㉿kali)-[/home/kali]
└─# sudo ip link set eth0 up

(kali㉿kali)-[/home/kali]
└─# ifconfig
docker0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255
    ether c2:8b:22:7c:ed:40 txqueuelen 0 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 5 overruns 0 carrier 0 collisions 0

eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.0.107 netmask 255.255.255.0 broadcast 192.168.0.255
    inet6 fe80::3d2c:9e04:ad76:6e67 prefixlen 64 scopeid 0<link>
    ether 00:0c:29:b6:28:b8 txqueuelen 1000 (Ethernet)
    RX packets 128 bytes 11897 (11.6 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 56 bytes 5792 (5.6 KiB)
    TX errors 0 dropped 2 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 16 bytes 1128 (1.1 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 16 bytes 1128 (1.1 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

- **Observation :** Temporary IPv6 address changed after reconnecting the network.

4. **Simulate a Zomato-style search:** Write a small script (any language) that stores an array of restaurant names and their IPv6 addresses, then allows the user to input a partial name and returns the matching restaurant's IPv6 address.

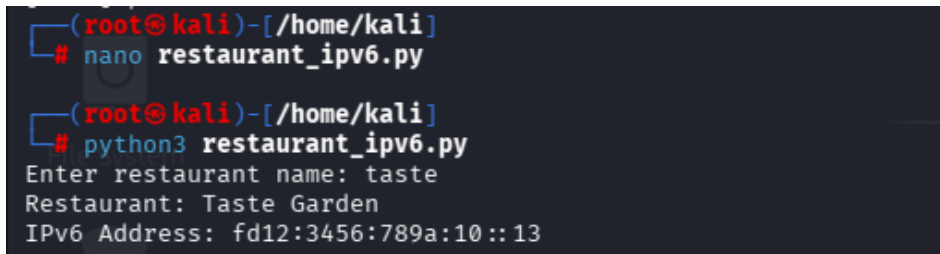
ANS:

- **Create file:**

- nano restaurant_ipv6.py

- **Write A Code**

```
restaurants = {  
    "Spicy Hub": "fd12:3456:789a:10::11",  
    "Food Plaza": "fd12:3456:789a:10::12",  
    "Taste Garden": "fd12:3456:789a:10::13",  
    "Urban Bites": "fd12:3456:789a:10::14",  
    "Royal Dine": "fd12:3456:789a:10::15"  
}  
  
query = input("Enter restaurant name: ").lower()  
  
found = False  
  
for name, ip in restaurants.items():  
    if query in name.lower():  
        print("Restaurant:", name)  
        print("IPv6 Address:", ip)  
        found = True  
  
if not found:  
    print("No matching restaurant found")
```



```
(root@kali)-[/home/kali]  
# nano restaurant_ipv6.py  
  
(root@kali)-[/home/kali]  
# python3 restaurant_ipv6.py  
Enter restaurant name: taste  
Restaurant: Taste Garden  
IPv6 Address: fd12:3456:789a:10::13
```

5. Use ChatGPT or Copilot to generate a sample IPv6 subnetting scenario (with at least 3 subnets), and then manually assign a valid IPv6 address from one of those subnets to your system. Paste the AI-generated scenario and your configuration steps in a text file.

ANS :

- **Create file:**
 `nano ipv6_subnetting.txt`
- **Paste:**
 IPv6 Subnetting Scenario

 Base Network:
 `2001:db8:5000::/48`

 Subnet 1:
 `2001:db8:5000:1::/64`

 Subnet 2:
 `2001:db8:5000:2::/64`

 Subnet 3:
 `2001:db8:5000:3::/64`

 Assigned Address:
 `2001:db8:5000:2::25/64`
- **Save and Exit:**
 CTRL + O
 ENTER
 CTRL + X
- **Assign Address from Subnet 2:**
 `sudo ip -6 addr add 2001:db8:5000:2::25/64 dev eth0`
- **Verify IPv6 Configuration:**
 `ip -6 addr show dev eth0`
- **Display the File:**
 `cat ipv6_subnetting.txt`



```
(root@kali)-[/home/kali]
# nano ipv6_subnetting.txt

(root@kali)-[/home/kali]
# sudo ip -6 addr add 2001:db8:5000:2::25/64 dev eth0

(root@kali)-[/home/kali]
# ip -6 addr show dev eth0
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    inet6 2001:db8:5000:2::25/64 scope global
        valid_lft forever preferred_lft forever
    inet6 fe80::3d2c:9e04:ad76:6e67/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
Trash
(root@kali)-[/home/kali]
# cat ipv6_subnetting.txt
IPv6 Subnetting Scenario

Base Network:
2001:db8:5000::/48

Subnet 1:
2001:db8:5000:1::/64

Subnet 2:
2001:db8:5000:2::/64

Subnet 3:
2001:db8:5000:3::/64

Assigned Address:
2001:db8:5000:2::25/64
```